

HDOC DAY-25

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Batch-15

Question:

Write a C program using nested loops, functions and switch statement to display the following decreasing patterns according to the user's choice:

1. Decreasing right-angled triangle using *
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

Aim -

To write a C program using nested loops, functions and switch statement to display different decreasing patterns selected by the user.

Algorithm

1. Start.
2. Display the menu containing four decreasing patterns.
3. Enter the user's choice.
4. Enter the number of rows n.
5. Use a switch statement to select the required pattern.
6. Call the corresponding function.
7. Use nested loops inside the function to print the selected pattern.
8. Stop.

>C program:

```
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day25.c
Untitled2

1 #include <stdio.h>
2
3 void pattern1(int n){
4     for(int i=1; i<=n; i++){
5         for(int j=1; j<=i; j++){
6             printf(" ");
7         }
8     }
9 }
10
11 void pattern2(int n){
12     for(int i=1; i<=n; i++){
13         for(int j=1; j<=i; j++){
14             printf(" ");
15         }
16     }
17 }
18
19 void pattern3(int n){
20     for(int i=1; i<=n; i++){
21         for(int j=1; j<=i; j++){
22             printf(" ");
23         }
24     }
25 }
26
27 void pattern4(int n){
28     for(int i=1; i<=n; i++){
29         for(int j=1; j<=i; j++){
30             printf(" ");
31         }
32     }
33 }
34
35 int main(){
36     int choice, n;
37     printf("1. Decreasing right-angled triangle\n");
38     printf("2. Decreasing natural-number pattern\n");
39     printf("3. Decreasing repeated-number pattern\n");
40     printf("4. Reverse-number triangle\n");
41     printf("Enter choice: ");
42     scanf("%d", &choice);
43     printf("Enter number of rows: ");
44     scanf("%d", &n);
45     switch(choice){
46         case 1: pattern1(n); break;
47         case 2: pattern2(n); break;
48         case 3: pattern3(n); break;
49         case 4: pattern4(n); break;
50         default: printf("Invalid choice"); }
51     return 0;
52 }
```

>OUTPUT EXECUTION:

```
Session Terminal Emulator
Use the command line
prityadarshna@kali: ~
prityadarshna@kali:~$ gcc day25.c -o day25
prityadarshna@kali:~$ ./day25
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter choice: 1
Enter number of rows: 4
* * * *
* * *
* *
*

prityadarshna@kali:~$ ./day25
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter choice: 2
Enter number of rows: 4
1 2 3 4
1 2 3
1 2
1

prityadarshna@kali:~$ ./day25
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter choice: 3
Enter number of rows: 4
4 4 4 4
3 3 3
2 2
1

prityadarshna@kali:~$ ./day25
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter choice: 4
Enter number of rows: 4
4 3 2 1
3 2 1
2 1
1

prityadarshna@kali:~$
```